A Human Cornea-on-a-Chip for the Study of Epithelial Wound Healing by Extracellular Vesicles*

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***Abstract*— We present a microfluidic platform to recapitulate human cornea's complexity and its interface with external environment. According to the cornea's anatomy, the fully integrated human cornea on the chip consists of human corneal cells and a porous membrane embedded in a microfluidic platform, replicating multi-scale structural organization and biological phenotypes. Using this system, we discover and verify the fully integrated human cornea barrier function on the chip. Besides, we develop an *in vitro* model of the corneal epithelial wound to study the cell-free therapy efficacy using extracellular vesicles (EVs). We find that EVs derived from bone marrow mesenchymal stem cells (BM-MSCs) can significantly accelerate corneal epithelial wound healing. A dose-dependent effect is also observed in the experiments. This work improves our ability to simulate the interaction between human cornea and external world *in vitro*. It may contribute to the future development of new screening platforms for biopharmaceuticals.**

I. Introduction

Human eye is the visual organ and the most important organ in human senses. As the main protective barrier in this unit, the cornea consists of epithelium, Bowman's layer, stroma, Descemet's layer, and endothelium. The ocular surface is continuously affected by dynamic microenvironment [1]. Exposure to the external environment, the human cornea is sensitive to environmental agents associated with eye damage and vulnerable to traumatic injuries, damaging the corneal barrier and even cause a loss of vision.

BM-MSCs have shown excellent research and practical value in ophthalmic diseases [2]. However, it remains uncertain whether the MSCs have long-term adverse effects on the immune system and whether there is a possibility of inducing tumorigenesis. On the other hand, the therapeutic effects of MSCs are widely mediated by MSCs differentiation and paracrine signaling via EVs [3]. The cell-free nature of EVs has gained particular interest concerning their safety on therapeutics [4]. Therefore, extracellular vesicles derived from bone marrow mesenchymal stem cells (MSC-EVs) have been widely studied for cell-free therapy applications. Thus, MSC-EVs could be of great potential for corneal wound healing.

Most ophthalmic corneal treatment studies are carried out through *in vitro* cell culture or animal model conditions. However, *in vitro* studies are usually carried out under static conditions different from the dynamic conditions *in vivo*, such as blinking and tear secretion [5]. Although animal models can reflect the organism's overall situation, studies have shown that these tests have limited ability to generalize the complex structure of the human cornea due to species differences. For example, the low blink rate of rabbits generates higher interactions between drug and the ocular surface, which leads to higher drug penetration and may mislead preclinical test results [6]. Moreover, the ocular surface's structural, functional, and environmental complexity pose major technical challenges for *in vitro* studies of its physiology and pathology using traditional cell culture models. These critical drawbacks of exiting models have limited our fundamental understanding and hindered new treatments for ocular diseases.

Here, we demonstrate a human cornea-on-a-chip, the microfluidic device's functionality is characterized and verified by building the human cornea model in this study. Furthermore, the microfluidic chip is used to establish an *in vitro* corneal injury model to study whether cell-free therapy promotes corneal wound healing. We find that MSC-EVs can significantly promote cell migration. We develop this novel micro-engineered device that improves the interaction between the *in vitro* and external environments, providing a potential solution to obtain predictive results to *in vivo* observations.

II. Methods

A. Device design and fabrication

The adult cornea consists of a stratified non-keratinizing epithelial cell layer, a thick, highly aligned collagenous stroma interspersed with keratocytes, and a single cell layered endothelium (Fig. 1 (a)). The epithelium sits atop a Bowman's layer, and the endothelial layer resides on a thickened basement membrane, termed Descemet's membrane. Here, we design a three-dimensional *in vitro* ocular surface model. A membrane modified with

*Resrach supported by Key-Area Research and Development Program of Guangdong Province (2019B020226004), National Natural Science Foundation of China (61805271, 62074155), Shenzhen Science and Technology Innovation Commission (JCYJ20170818154035069, KCXFZ202002011008124).

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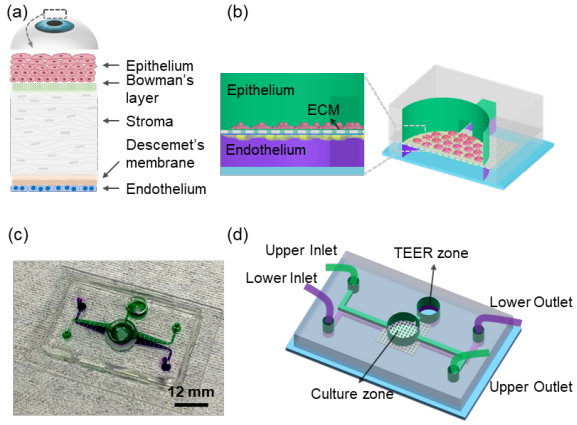

Fig. 1. Schematic diagram of human cornea and design for the human cornea-on-a-chip. (a) The anatomy of the human cornea. (b) Cross-sectional schematic diagram of the corneal chip. HCEpi cells and HCEnd cells are cultured on the opposite sides of the membrane, the ECM-coated porous membrane is embedded in between the two PDMS layers. (c) Photograph of the human cornea-chip, upper and lower microfluidic structures are indicated by using solutions with purple and green color dyes. (d) The cornea-chip with two individual microfluidic channels and a circular hole at the center mimicking the human corneal structure. Structures for in situ TEER measurements are also fabricated on the chip.

extracellular matrix (ECM) is used to simulate the middle three layers, immortalized human corneal epithelial (HCEpi) cells and endothelial (HCEnd) cells are cultured on each side of the membrane by introducing through different microchannels (Fig. 1 (b)). The open top of the cell culture area provides an air-liquid interface for the corneal epithelium, so the device's physiological characteristics are similar to human cornea. A hole for trans-epithelial electrical resistance (TEER) measurements *in situ* is also made on the device and connected to the cell culture area that is located at the middle of the microfluidic channels (Fig. 1 (d)).

The microfluidic device is fabricated by directly bonding a glass substrate with two polydimethylsiloxane (PDMS) layers sequentially, and an ECM-coated polycarbonate (PC) membrane is sandwiched between the two PDMS layers. The upper and lower PDMS layers with microchannels are replicated from SU-8 structures fabricated by soft lithography. A 10:1 (w/w) mixture of PDMS prepolymer and curing agent is cast over the SU-8 molds and cured at 80 °C overnight. The PDMS replicas are peeled off from the molds, resulting in microfluidic channels with a height of 220 µm. Access holes for inlets and outlets of the microfluidic channels are punched on the PDMS slabs. At the middle of the microfluidic channel, a circular hole with a diameter of 6 mm is punched on each PDMS slab. This is used to mimic the human corneal structure, and the open-top surface of this area is used to create an air-liquid interface and mimic the ocular surface more precisely. By thoroughly cleaning the glass substrate and the PDMS layers, they are then bonded together by oxygen plasma with the PC membrane embedded between the two microfluidic channels. Details on the microfabrication procedure are presented in Fig. 2.

B. Human cornea-chip construction and characterization

Before the surface treatment, the chip is sterilized by UV germicidal irradiation in a biosafety cabinet for 30 minutes. Then, collagen (0.05 mg/mL in 0.02 M acetic acid) is used as the ECM to treat the PC membrane at 37 °C for 2 hours. Excess proteins are then aspirated, and the membrane is allowed to air dry. HCEpi cells and HCEnd cells are introduced into the device at a concentration of 3×10^6 cells/mL and 5×10^5 cells/mL through the upper and lower microchannels, respectively. The cells are then seeded on each side of the membrane and cultured by their medium at a continuous flow condition with a volumetric flow rate of 100 µL/h. Using a programmable syringe pump, the device is kept in a CO_2 incubator. Once the epithelial layer is confluent on the membrane, the culture medium is gently aspirated from the 6 mm diameter hole at the middle of the upper microfluidic channel. The air is then filled in the epithelial compartment to create an air-liquid interface and mimic the ocular surface. As a comparison, the HCEpi cells are also cultured in a transwell chamber with an air-liquid interface, and details are described in Fig. 4 (a). TEER values are measured *in situ* using a Millicell-ERS electrical resistance system on different days to evaluate the corneal epithelium's barrier integrity. After 14 days, the cornea-chip is characterized using immunofluorescence staining and monitored by a Zeiss Axio Observer 7 inverted fluorescence microscope equipped with light-emitting diodes, appropriate

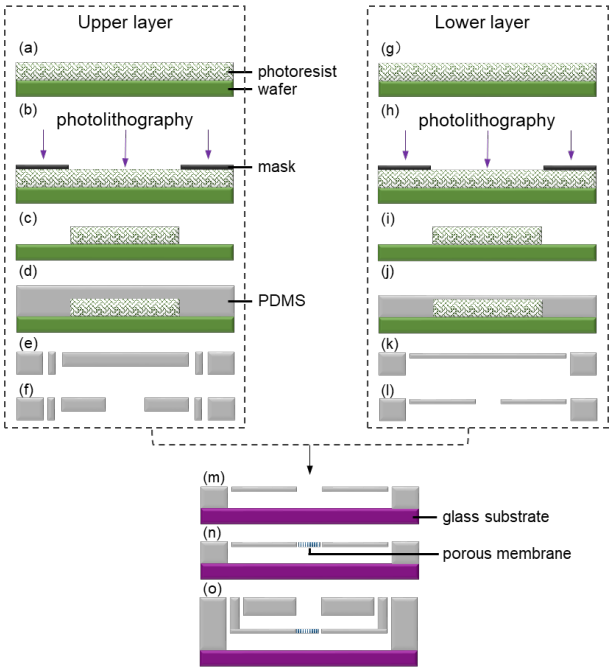

Fig. 2. Schematic illustration on microfabrication of the cornea-chip. (a), (g) SU8 photoresist is spin-coated on the silicon wafers and (b), (h) patterned by UV exposure and (c), (i) developed. (d), (j) PDMS is poured onto the SU8 molds. (e), (k) The PDMS slabs are peeled off and punched to make access holes. (f), (l) The PDMS slabs are further punched to make open wells. (m) The lower PDMS slab and the glass substrate are bonded by oxygen plasma treatment. (n) The open well in the lower PDMS slab is sealed by a porous polycarbonate membrane. (o) The upper PDMS slab is bonded to the lower layer to create a multilayered cornea chip.

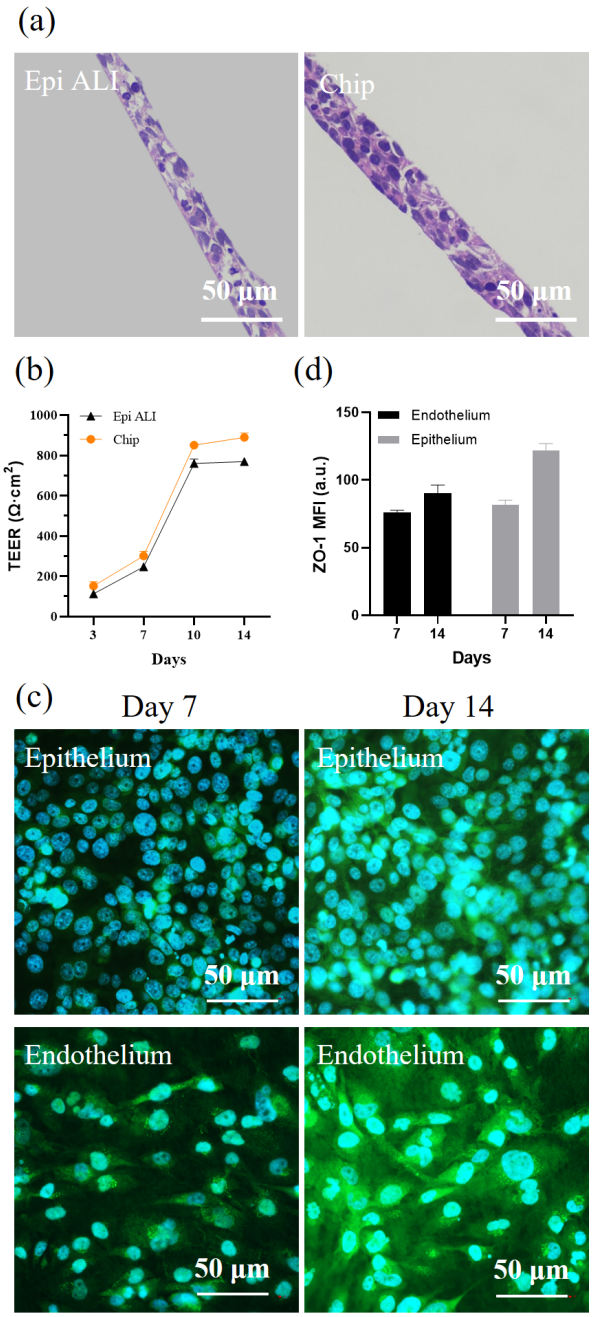

Fig.3. Construction of physiological phenotypes of the human cornea in the device. (a) Representative hematoxylin and eosin (H&E) images of the stratified corneal epithelium for 14 days. Epi ALI, HCEpi cells cultured at air-liquid interface in transwell inserts. Chip, HCEpi cells cultured at air-liquid interface in cornea chip. (b) TEER values of the Epi ALI and the cornea chip. (c) Immunofluorescent characterization of HCEpi and HCEnd cells co-cultured in cornea-chip. ZO-1 protein, green color. Cell nucleus, blue color. (d) Mean fluorescent intensity of ZO-1 protein expression obtained from different groups in (c). MFI, mean fluorescence intensity. a.u., arbitrary unit. Data are mean ± S.D. of relatively fluorescence intensity from three independent experiments.

fluorescent filter sets, and a sCMOS camera (Hamamatsu ORCA-Flash4.0-V3).

C. *Scratch wound healing study in the cornea-chip*

A scratch wound is made on the epithelium layer by a pipet tip (Fig. 4 (a)). Scraped cells are removed by rinsing with PBS, and the cultures are photographed at time 0 using a light microscope (Zeiss Axio Observer 7). The wound model is then treated by serum-free cell culture medium as the control group and 1×10^8 MSC-EVs/mL, respectively. More details on the amount of MSC-EVs are provided in Fig. 4 (b). After 24 hours, the wounds treated with different dosages are again photographed, and the margins are traced. The area is quantified using ImagePro Plus and compared to the wound area at time 0 (data not shown). In addition, 10% Cell Counting Kit-8 (CCK-8) agent is added to the medium, and then cells are incubated for another 3 hours at 37 °C in the dark. Absorbance at 450 nm is measured with a Microplate Reader, and the results are normalized to the control group.

D. *Statistical analysis*

All results are presented as mean ± standard deviation (S.D.). The unpaired t-test or two-way analysis of variance (ANOVA) followed by Bonferroni's multiple comparisons test is performed using GraphPad Prism 8. The experiments are repeated and P values are obtained from source data and reported in figure legends.

III. Resluts and Discussion

To investigate whether the chip is suitable for human corneal cells culture, the cells are maintained either on a transwell chamber with an air-liquid interface ("Epi ALI" in Fig. 3 (a)) for comparison, or in the microfluidic device for 14 days on the ECM-coated PC membrane with an air-liquid interface ("Chip"), respectively. For constructing the biomimetic human corneal tissue in our device, the seeded epithelial cells are first grown to confluence in submerged conditions and then cultured at the air-liquid interface to mimic their native organ-specific environment. The HCEpi cells are cultured on the microfluidic chip form a stratified epithelium consisting of three to five stacked cell layers, which recapitulate the corneal epithelium *in vivo* (Fig. 3 (a)). The TEER is measured between the HCEpi and the HCEnd layer to evaluate barrier integrity after culturing cells on chip for 3, 7, 10, and 14 days, respectively (Fig. 3 (b)). The measurement results are presented as $\Omega \cdot cm^2$ by multiplying the resistance with the surface area (0.28 cm^2) of the culture zone. The resistance of the unseeded cell membrane is subtracted from all readings before calculations. The human corneal epithelium begins to form a barrier function when the cells are cultured under the air-liquid interface condition for 3 days. After culturing for a long time, the barrier function obtained from the cornea-chip is higher than that from the transwell model.

As the epithelial and endothelial tight junctional barrier plays a crucial role in homeostasis and host defence of cornea [7], the expression of tight junction proteins can reflect the barrier function. A substantial increase in the expression of zona occludens-1 (ZO-1) protein accompanies the full barrier function. The HCEnd or HCEpi cells are co-cultured in the chip with an air-liquid interface, and the cell morphology and the expression of ZO-1 of those cells are shown in Fig. 3 (c). After 14 days, the cells' morphology remains very well, and the expression of ZO-1 is

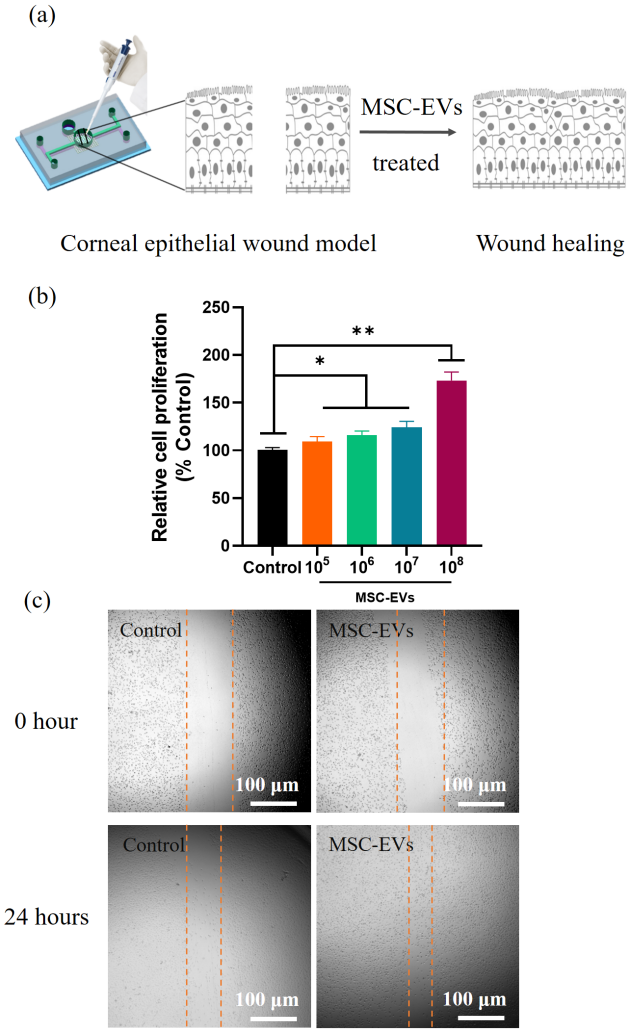

Fig. 4. MSC-EVs stimulate scratch wound healing of immortalized human corneal epithelial cells on-chip. (a) Schematic diagram of the corneal epithelial wound healing model. (b) The dose response of MSC-EVs on epithelial wound healing and proliferation. CCK-8 assay showed increased proliferation of HCEpi cells that were incubated with MSC-EVs after 24 hours. (c) Representative experiment showing wound closure over time. The cells are treated by culture medium ("Control"), EVs derived from MSC ("MSC-EVs"). The approximate wound edges are indicated with orange lines. All data from three independent experiments are normalized to the control group. "*" P<0.05, "**" P<0.01.

significantly increased on both HCEnd and HCEpi cells that are cultured on the chip (Fig. 3 (d)).

A scratch wound is made on the epithelium layer by a pipet tip (Fig. 4 (a)). Scraped cells are removed by rinsing with PBS. The corneal epithelial wound is then treated with different MSC-EVs concentrations (1×10^5, 1×10^6, 1×10^7, or 1×10^8 MSC-EVs/mL) for 24 h (Fig. 4(b)). The CCK-8 results show that HCEpi cells' proliferation is increased by incubating with MSC-EVs for 24 hours. A dose-dependent effect is also observed in the experiments. The cell proliferation increases with the concentration of MSC-EVs, indicating that the concentration of 1×10^8 MSC-EVs/mL can generate a better wound healing effect. Such concentration is used to treat the scratch wound model for further research.

The approximate wound edges at 0 and 24 hours are indicated with yellow lines in Fig. 4(c). The area between the leading edges of the wound is determined at 24 hours and subtracted from the wound area at time 0 to obtain a value for the wound size difference. The results obtained from the experimental groups are normalized by that of the control group. MSC-EVs enhance wound healing by approximately 150%, demonstrating that the MSC-EVs are more conducive to cell migration.

IV. Conclusion

We develop a human cornea-on-a-chip using microfluidic technology in this study, validate its barrier effects, and test a cell-free therapeutic method to heal a scratch wound using MSC-EVs. In brief, the cornea-chip has a complete human corneal structure, and its physiological characteristics are much closer to that of the human cornea, so it is suitable for corneal disease study and evaluation of ocular drug metabolism in the future. This corneal chip may replace animal models for cornea-related testing, which will provide meaningful, predictive, and complimentary results to *in vivo* and *ex vivo* observations. Therefore, we believe that our technique could evolve into a tool for a comprehensive class of biological and clinical applications.

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